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Analytical researchers used cation exchange chromatography to separate and quantify the FAAs in various samples of Grana Padano cheese. First, they prepared standards containing the 20 proteinogenic amino acids with the pH adjusted so that all amino acids were fully protonated. The standards were then run through a cation exchange column to calibrate amino acid separation and detection by the instrument. FAAs were then extracted from the cheese samples, prepared in the same way as the standards, and run on the same column. An example chromatogram is shown in Figure 1.

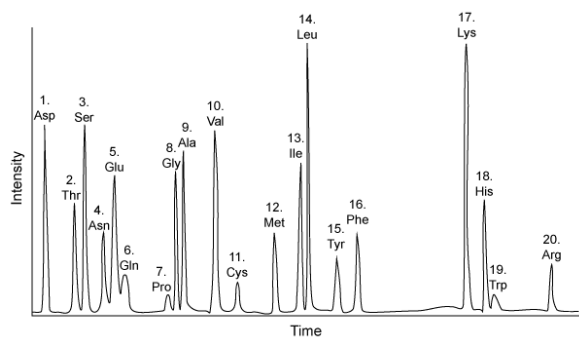


Figure 1 Chromatogram of the FAA content obtained from a sample of Grana Padano cheese.

To confirm the identities of several FAAs with ionizable side chains, fractions collected from the column at the appropriate times were acidified and titrated with 1.0 M NaOH. The pK_a values of the N- and C-termini and several side chains are shown in Table 1.

Table 1 pK_a values for various amino acid chemical groups. Note: N- and C-terminal pK_a values are averages of the values from all 20 proteinogenic FAAs.

Chemical Group	pK_a
N-terminus	9.5
C-terminus	2.2
Arginine side chain	12.5
Lysine side chain	10.5
Tyrosine side chain	10.0
Cysteine side chain	8.0
Histidine side chain	6.0
Glutamic acid side chain	4.3
Aspartic acid side chain	3.7

The pH of the FAA solution was most likely adjusted to which of the following ranges prior to loading onto the column?

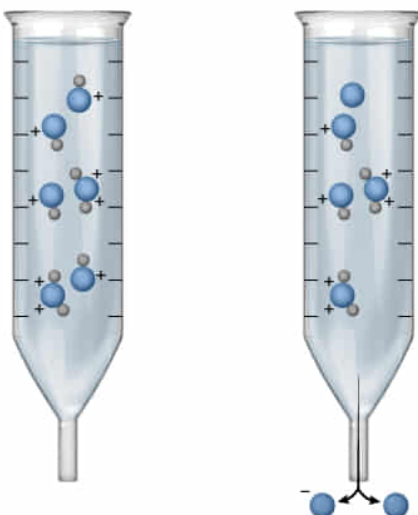
- ✓ ☒ A. pH < 2.2 [62%]
☐ B. 2.2 < pH < 7.0 [18%]
☐ C. 7.0 < pH < 12.5 [12%]
☐ D. pH > 12.5 [6%]

Correct

61% answered correctly

Explanation:

Cation exchange and pH



When pH < 2.2, all amino acids have a charge of +1 or +2 and bind the column

When pH > 2.2, some amino acids are uncharged or negatively charged and do not bind the column

Cation exchange columns are chromatography columns containing resins with negatively charged residues. These residues bind positively charged molecules (cations). Once the cations are bound, the buffer running through the column is gradually changed by increasing the pH and/or salt concentrations. **As the pH rises, protons are removed** from the bound molecules, causing the molecules to **lose positive charge** and **eventually elute** (ie, come off the column). Similarly, cations in the **salts added to the column compete** with the bound cationic molecules, **displacing them from the column** (ie, cations are exchanged).

To bind the column in the experiment, all the amino acids must initially have a net positive charge. This can be accomplished by fully protonating all amino acids, which in turn is accomplished by bringing the pH below the lowest pK_a of all the amino acids present. Table 1 shows that the lowest pK_a of any chemical group in the 20 **proteinogenic amino acids** is that of the C-terminus (2.2). Therefore, for all the amino acids to be cations, the solution must be adjusted so that $pH < 2.2$.

(Choices B and C) pH levels between 2.2 and 12.5 would cause some amino acids to be neutral or negatively charged at the start of the experiment, and they would not bind the column. For effective separation, all amino acids must bind the column and *then* be selectively eluted, one at a time.

(Choice D) An initial pH greater than 12.5 would cause all amino acids in solution to have a net *negative* charge. This would be ideal for an **anion exchange column**, not a cation exchange column.

Educational objective:

Cation exchange columns bind cationic molecules. These columns are often used with a low initial pH to ensure full protonation of all molecules. The pH may then be gradually increased (along with salt concentrations) to allow elution of the bound molecules.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403181

System:
Amino Acids and Proteins

In addition to amino acids incorporated into proteins, many foods contain free amino acids (FAAs). In protein-rich foods that require a ripening or fermentation process, such as cheeses, proteins are often degraded, causing FAAs to constitute as much as 25% of the total amino acid content of the food. As such, the FAA profiles of different cheeses may affect their taste and nutritional qualities.

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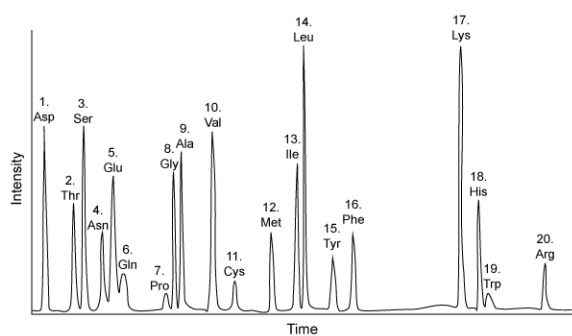


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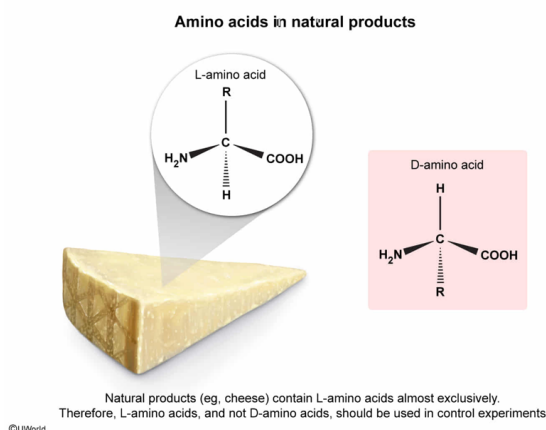
With the exception of glycine, all the amino acids used in the standard solution to calibrate the column should have been:

- ☐ A. R-amino acids [4%]
- ☐ B. S-amino acids [10%]
- ☐ C. D-amino acids [9%]
- ✓ ☒ D. L-amino acids [75%]

Correct

75% answered correctly

Explanation:



The 20 **proteinogenic amino acids** (ie, amino acids that are commonly found in proteins) are **chiral molecules**, except for **glycine**. The chiral amino acids are designated as **L- or D-amino acids**, depending on the orientation of the substituents around the α -carbon (the chiral center). The **chiral amino acids produced by living organisms are almost exclusively L-amino acids**.

The researchers used amino acid standards (ie, an amino acid solution of known composition) to calibrate their chromatography instruments, allowing them to later identify the amino acids found in cheese samples. To ensure that the peaks on the standard and sample chromatograms appear in the same positions, the standard should use identical amino acids to those found in the cheese samples. Therefore, with the exception of glycine, all the **standard amino acids used** should be **L-amino acids**.

(Choices A and B) Most proteinogenic L-amino acids have an S-**absolute configuration** at the **α -carbon**. However, L-cysteine has an R-configuration at the alpha carbon, so the researchers should use cysteine in the R-configuration, but the other amino acids should have an S-configuration.

(Choice C) With few exceptions, D-amino acids are not found in nature and should not be used for the calibration standard.

Educational objective:

Living organisms produce L-amino acids almost exclusively. D-amino acids are rarely found in natural samples.

Time spent:0 sec

Subject:
Biochemistry

QID:403182

System:
Amino Acids and Proteins

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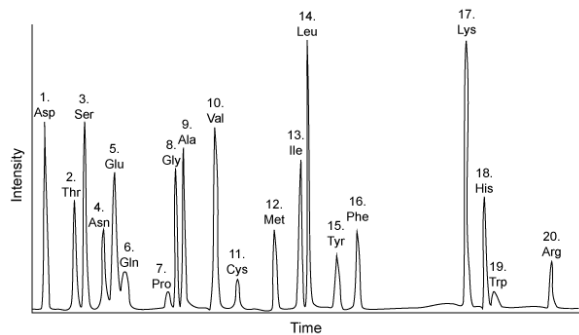


Figure 1 Chromatogram of the FAA content obtained from a sample of Grana Padano cheese.

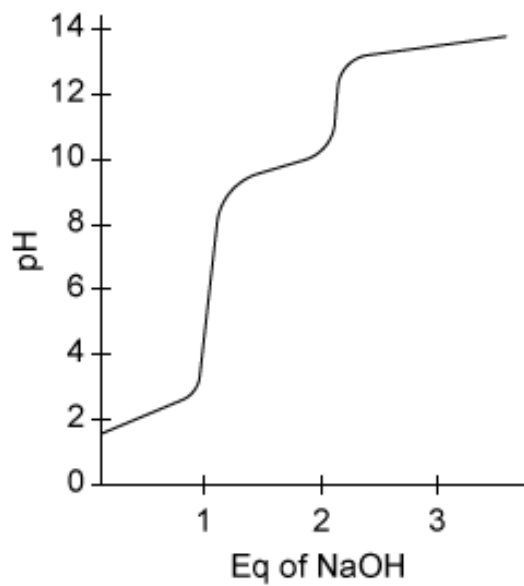
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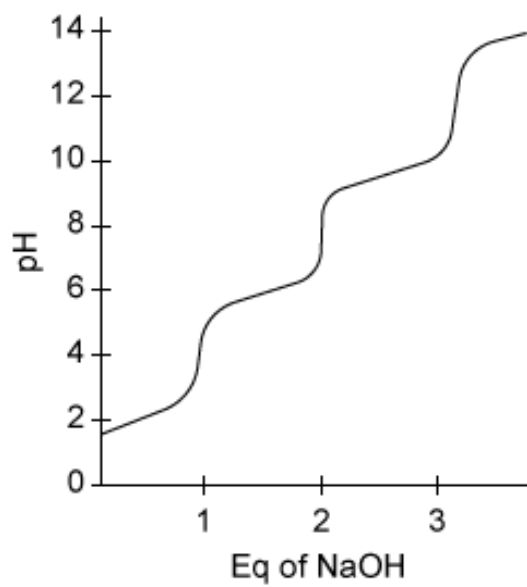
Which of the following titration curves corresponds to the amino acid found in peak 17 of the chromatogram?

☐ A.



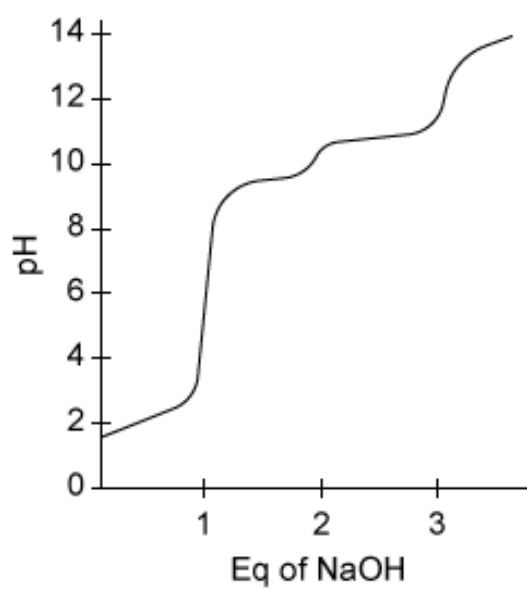
[29%]

☐ B.



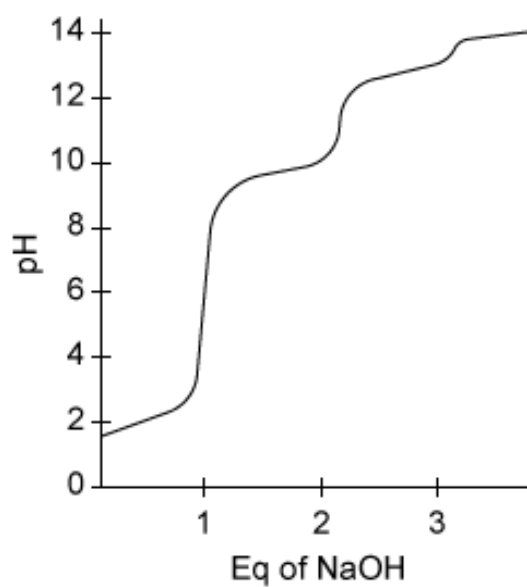
[18%]

✓ ☒ C.



[39%]

☐ D.

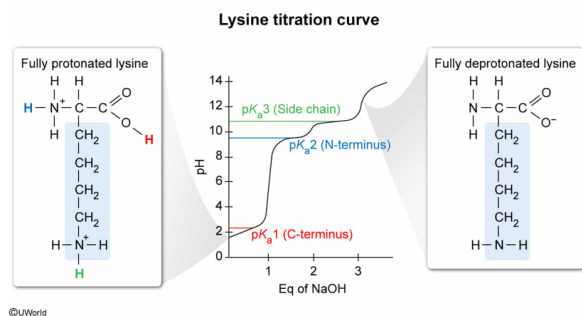


[13%]

Correct

38% answered correctly

Explanation:



Of the 20 **proteinogenic amino acids**, 7 have side chains that can be ionized at certain pH levels between 0 and 14. These 7 amino acids (R, K, Y, C, H, E, D) are collectively referred to as the **ionizable amino acids**. Each ionizable amino acid has a characteristic **titration curve**, with **buffer regions** surrounding the pK_a s of the C-terminus, the N-terminus, and the side chain.

Figure 1 identifies peak 17 as Lys (ie, lysine). **Table 1** shows that the side chain of lysine has a pK_a of 10.5, and that the C- and N-termini have pK_a s of 2.2 and 9.5, respectively. Therefore, the titration curve should show buffer regions surrounding the pH levels of 2.2, 9.5, and 10.5. It will also show a relatively flat region at the end of the titration as the pH approaches 14 because this is the pH of the 1.0 M NaOH used in the titration, and the pH of the solution never exceeds this limit. Only the graph depicted in **Choice C** correctly shows all three buffer regions.

(Choice A) This graph shows the titration curve of a nonionizable amino acid, with only **2 buffer regions** (ie, 2 pK_a s).

(Choice B) This graph shows the **titration curve of histidine**, with a buffer region surrounding pH 6.0. Histidine was found in peak 18, not peak 17.

(Choice D) This graph shows the **titration curve of arginine** (peak 20), with a buffer region surrounding pH 12.5.

Educational objective:

The ionizable amino acids have characteristic titration curves with buffer regions corresponding to their N-terminal, C-terminal, and side chain pK_a s.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403183

System:
Amino Acids and Proteins

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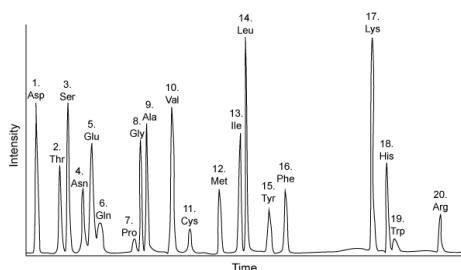


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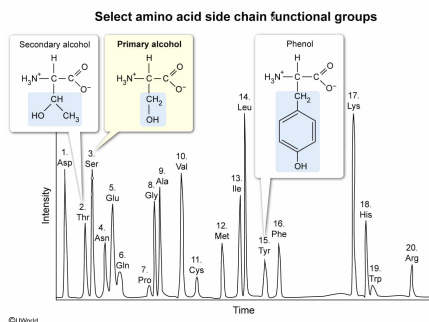
Which peak(s) in Figure 1 correspond to amino acids that have primary alcohols in their side chains?

- ☐ A. Peak 2 only [5%]
- ✓ ☒ B. Peak 3 only [52%]
- ☐ C. Peaks 2 and 3 only [16%]
- ☐ D. Peaks 2, 3, and 15 only [25%]

Correct

52% answered correctly

Explanation:



The identity of an amino acid is determined by the structure of its side chain. Each side chain can be characterized by the **functional groups** it contains, and by the chemical and physical properties conferred by those functional groups.

The question asks which peak or peaks correspond to amino acids with **primary alcohols** in their side chains. A primary alcohol is a hydroxyl (-OH) group bonded to a carbon atom, where the carbon atom is in turn bonded to two hydrogen atoms and another carbon. Five of the **proteinogenic amino acids** contain hydroxyl groups in their side chains when fully protonated: aspartic acid (Asp), glutamic acid (Glu), serine (Ser), threonine (Thr), and tyrosine (Tyr).

However, the hydroxyl groups in **Asp** and **Glu** are part of **carboxylic acid** groups, and therefore they are not alcohols. The hydroxyl group in **Tyr** is linked to a benzene ring, making the functional group a **phenol**, which is not generally considered an alcohol. **Thr** contains an alcohol group, but it is a **secondary alcohol** because the carbon to which the hydroxyl group is bound is linked to *two* other carbons instead of one. Therefore, only **Ser** contains a primary alcohol in its side chain. Serine is represented by **peak 3** on the chromatogram.

(Choices A, C, and D) Peaks 2 and 15 correspond to threonine (a secondary alcohol) and tyrosine (a phenol), respectively. They do *not* contain primary alcohols.

Educational objective:

Each amino acid has a unique side chain that can be characterized by its functional groups. Serine and threonine have primary and secondary alcohols in their side chains, respectively.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403184
System:
Amino Acids and Proteins